

FARMINGDALE REPUBLIC, NY, USA

WMO: 744864

Lat: 40.734N

Lon: 73.417W

Elev: 25

StdP: 101.03

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.6	-21.2	0.6	-9.2	-18.6	0.7	-6.6	12.2	2.8	11.2	1.6	4.6	310	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	32.1	23.2	30.2	22.3	28.7	21.8	24.9	29.0	24.2	27.7	23.5	26.8	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	26.8	23.0	17.7	25.9	22.5	17.2	25.5	75.6	29.0	72.8	27.9	70.0	26.5	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.9	9.3	8.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-13.7	34.8	2.7	2.1	-15.7	36.3	-17.2	37.5	-18.8	38.7	-20.7	40.2
			-14.9	26.3	2.4	0.9	-16.7	27.0	-18.1	27.5	-19.5	28.0	-21.3	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	0.1	1.3	4.7	10.3	15.6	20.7	24.1	23.5	19.8	13.7	7.9	3.3	(6)
(7)		DBStd	9.26	5.32	4.54	4.49	3.80	3.63	3.50	2.70	2.66	3.26	4.14	4.24	4.57	(7)
(8)		HDD10.0	1082	309	244	172	41	1	0	0	0	13	89	213		(8)
(9)		HDD18.3	2782	566	477	422	244	103	17	0	1	22	152	314	467	(9)
(10)		CDD10.0	1867	1	1	9	48	174	322	438	420	295	129	26	5	(10)
(11)		CDD18.3	526	0	0	0	2	17	88	180	162	67	9	0	0	(11)
(12)		CDH23.3	3454	0	0	2	19	110	546	1370	1089	289	29	0	0	(12)
(13)		CDH26.7	926	0	0	0	4	21	147	423	282	44	4	0	0	(13)
(14)	Wind	WSAvg	4.0	4.3	4.4	4.7	4.4	3.9	3.7	3.7	3.4	3.6	3.9	4.0	4.2	(14)
(15)	Precipitation	PrecAvg	1123	80	81	109	93	89	101	90	106	82	91	90	105	(15)
(16)		PrecMax	1280	125	146	212	166	124	244	151	286	202	394	132	166	(16)
(17)		PrecMin	891	48	31	41	38	31	29	38	42	27	0	34	51	(17)
(18)		PrecStd	137	22	34	47	39	27	61	31	58	47	86	28	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.7	20.4	26.1	29.1	32.6	35.0	33.2	30.3	26.7	19.7	16.4	(19)
(20)		MCWB	11.1	10.7	13.3	16.0	20.0	23.1	24.1	24.2	22.4	19.5	14.8	13.3		(20)
(21)		2%	DB	12.0	12.1	16.0	21.2	26.4	30.2	32.5	31.2	27.7	23.2	17.5	13.8	(21)
(22)		MCWB	10.4	8.5	10.2	13.6	18.8	21.9	23.3	23.0	20.9	19.0	14.3	12.3		(22)
(23)		5%	DB	9.9	9.9	13.1	18.5	23.8	28.1	30.6	29.4	26.3	21.9	16.2	12.2	(23)
(24)		MCWB	7.8	7.4	9.2	12.1	17.6	21.1	22.8	22.0	20.5	18.1	13.8	10.9		(24)
(25)		10%	DB	7.8	7.9	11.2	16.3	21.7	26.4	28.9	27.9	24.9	20.0	14.6	10.5	(25)
(26)		MCWB	5.6	5.3	7.8	10.8	16.3	20.1	22.1	21.8	20.2	16.8	12.1	8.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.1	14.7	18.0	22.0	24.8	26.0	26.0	24.4	21.9	17.2	14.6	(27)
(28)		MDCB	13.1	13.5	19.7	23.7	27.6	30.2	32.1	30.8	27.3	23.6	18.4	15.5		(28)
(29)		2%	WB	10.7	9.8	11.9	15.4	20.3	23.5	24.9	25.0	23.4	20.4	15.7	12.6	(29)
(30)		MDCB	11.5	11.3	14.0	18.9	24.2	28.0	29.6	28.5	25.8	22.5	17.0	13.4		(30)
(31)		5%	WB	8.1	7.7	10.1	13.7	19.0	22.5	24.2	24.2	22.5	19.0	14.3	11.2	(31)
(32)		MDCB	9.3	9.0	12.4	16.3	22.5	26.4	28.2	27.2	24.7	21.0	15.6	12.2		(32)
(33)		10%	WB	5.9	5.7	8.2	12.2	17.6	21.5	23.6	23.5	21.6	17.6	12.7	8.7	(33)
(34)		MDCB	7.6	7.8	10.5	14.7	20.3	24.8	27.2	26.3	23.8	19.4	14.3	10.0		(34)
(35)	Mean Daily Temperature Range	MDBR	7.1	7.8	8.0	8.6	8.4	8.1	7.9	7.5	7.8	7.9	7.6	6.8		(35)
(36)		5% DB	MCDBR	9.3	10.1	10.7	12.0	11.5	10.1	9.6	8.8	8.5	9.1	8.6	8.4	(36)
(37)		MCWBR	8.3	8.0	7.0	6.5	6.1	4.8	3.8	3.7	4.5	5.6	7.0	8.4	(37)	
(38)		5% WB	MCDBR	9.0	9.8	10.0	9.8	10.1	8.8	7.9	7.0	6.6	7.5	7.7	8.0	(38)
(39)		MCWBR	8.9	9.1	7.8	6.7	6.1	4.9	3.9	3.7	4.5	5.8	7.3	8.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.319	0.344	0.382	0.430	0.480	0.489	0.471	0.399	0.351	0.323	0.311	(40)	
(41)		taud	2.503	2.483	2.459	2.379	2.274	2.173	2.158	2.203	2.377	2.524	2.554	2.553	(41)	
(42)		Ebn at Noon	853	898	909	891	851	806	793	797	838	850	833	825	(42)	
(43)		Edh at Noon	75	88	101	117	133	147	148	138	108	84	71	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.86	2.76	3.80	4.87	5.43	5.96	6.03	5.38	4.39	3.04	2.12	1.59	(44)	
(45)		RadStd	0.15	0.27	0.32	0.48	0.54	0.56	0.30	0.34	0.36	0.29	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+64

Nomenclature: See separate page